X-Fusion: Tri-Modal Transformer Fusion for Multi-Label Chest X-ray Analysis with Evaluation Integrity Analysis

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Abstract. Multi-label chest radiograph analysis benefits from integrating visual, textual, and structured clinical data, yet most existing systems evaluate multimodal fusion without verifying the independence of auxiliary inputs from target labels. We present X-Fusion, a tri-modal Transformer architecture that fuses BiomedCLIP image representations, CXR-BERT clinical-text embeddings, and six structured patient attributes in a shared 512-dimensional space through a learned classification token. Training combines Asymmetric Loss with image–text contrastive alignment for fourteen thoracic findings on patient-disjoint ChestX-ray14 subsets. We contribute three results: (i) a compact, reproducible multimodal architecture with an inference web interface exposing visual attention and metadata attribution; (ii) a systematic diagnosis showing that label-conditioned auxiliary inputs (generated notes and synthetic vital signs) inflate evaluation scores and invalidate clinical-performance claims; and (iii) a new independent-input reference experiment using only real image pixels and recorded demographics, with validation-selected hyperparameters and complete per-case prediction archives. The independent reference achieves macro AUROC 0.634 on 7,100 test images using logistic regression on 16×16 pixels and age/sex, establishing a reproducible lower bound. Our analysis demonstrates that evaluation integrity verification is essential before multimodal medical AI systems can claim clinical benefit.

Keywords: Chest radiography · Multimodal learning · Transformer fusion · Evaluation integrity · Reproducibility

1 Introduction

Chest radiography remains the most frequently performed diagnostic imaging examination worldwide, with a single radiograph potentially exhibiting multiple coexisting thoracic abnormalities [11]. Multi-label classification, where each image may receive several non-exclusive disease labels simultaneously, better reflects clinical reality than mutually exclusive single-diagnosis prediction. NIH ChestX-ray14 provides a widely adopted benchmark containing 112,120 frontal-view radiographs with fourteen pathology labels [11,7].

Clinical decision-making relies not only on the radiographic image but also on patient history, vital signs, and laboratory findings. Multimodal fusion architectures that

combine imaging with structured and unstructured clinical data have therefore attracted growing research interest [12,1]. However, a critical but often overlooked requirement is that auxiliary inputs must be *collected independently* of the reference diagnosis; otherwise, the evaluation conflates input information with target leakage.

We present X-Fusion, a tri-modal Transformer prototype that integrates radiographic, textual, and structured patient information for fourteen-label chest X-ray analysis. Beyond the architectural contribution, this paper makes three distinct contributions:

1. **Architecture and implementation.** A compact four-token fusion mechanism that combines pretrained BiomedCLIP visual features, projected CXR-BERT text representations, and an MLP-encoded metadata vector through a two-layer, eight-head Transformer encoder. An end-to-end web interface exposes predictions alongside visual attention overlays and metadata attribution views.
2. **Evaluation integrity analysis.** A systematic diagnosis of target leakage in the evaluation pipeline: the dataset generates clinical notes from disease labels and samples vital signs conditional on disease presence. We demonstrate that patient-disjoint splitting cannot remedy this within-case dependency and that the resulting near-perfect scores (macro AUROC ≈ 1.0) characterize the leakage pathway rather than clinical discrimination.
3. **Independent-input reference experiment.** New reference models trained exclusively on real image pixels and recorded demographics (age, sex), with no access to generated notes, synthetic vitals, or the original checkpoint. These provide reproducible baselines with complete experimental archives including per-case predictions, split manifests, and software version records.

The paper is organized as follows. Section 2 reviews related work. Section 3 describes the architecture and training methodology. Section 4 details the dataset and experimental setup. Section 5 presents the artifact reanalysis, independent baselines, and discussion. Section 6 addresses limitations and future directions. Section 7 concludes.

2 Related Work

Chest radiograph benchmarks. Wang et al. introduced ChestX-ray14 with weakly supervised labels extracted from radiology reports [11]. CheXpert addressed uncertain labels through expert comparison and demonstrated the impact of label uncertainty on model evaluation [5]. Both benchmarks use image-level labels without paired clinical context, making auxiliary-input integrity a concern for multimodal extensions.

Vision Transformers in medical imaging. The Transformer architecture [10] was adapted for image classification by Dosovitskiy et al. [3]. DAViT applied domain-adapted Transformers for single-disease pneumonia detection [4]. Our work extends this paradigm to multi-label, multimodal fusion.

Pretrained biomedical encoders. BiomedCLIP learns joint visual–textual representations from 15 million biomedical image–text pairs [12]. CXR-BERT specializes BERT for radiology text understanding [1]. X-Fusion uses both encoders as frozen or partially trainable feature extractors, projecting their outputs into a shared embedding space.

Training objectives for imbalanced multi-label tasks. Asymmetric Loss [9] down-weights easy negatives through asymmetric probability margins, addressing severe class imbalance in multi-label settings. InfoNCE [8] enables contrastive alignment between paired modalities.

Explainability methods. SHAP [6] provides model-agnostic feature attribution. Chefer et al. [2] showed that raw attention weights alone do not constitute validated relevance or localization evidence.

3 Proposed Architecture and Methodology

3.1 Modality-Specific Encoders

Given an RGB radiograph I, a tokenized clinical note T, and an ordered metadata vector $m \in \mathbb{R}^6$, three modality-specific encoders produce aligned 512-dimensional representations:

$$z_v = f_v(I) \in \mathbb{R}^{512}, \tag{1}$$

$$z_t = W_t\, h_t(T) + b_t \in \mathbb{R}^{512}, \tag{2}$$

$$u = \mathrm{Dropout}_{0.1}(\mathrm{GELU}(W_1 m + b_1)), \tag{3}$$

$$z_m = \mathrm{GELU}(W_2 u + b_2) \in \mathbb{R}^{512}. \tag{4}$$

Vision encoder. BiomedCLIP produces a pooled 512-dimensional image feature vector from the 224×224 input. **Text encoder.** CXR-BERT's first output token (768-d) is linearly projected to 512 dimensions; notes are limited to 256 tokens. **Metadata encoder.** A two-layer MLP maps six fields (age, sex, temperature, heart rate, respiratory rate, SpO_2) through dimensions $6 \to 128 \to 512$ with GELU activation and dropout.

3.2 Transformer Fusion

The three modality vectors join a learned classification token $e_{\mathrm{CLS}} \in \mathbb{R}^{512}$ to form a four-token input sequence:

$$S_0 = [e_{\mathrm{CLS}};\, z_v;\, z_t;\, z_m] \in \mathbb{R}^{4\times 512}. \tag{5}$$

Two Transformer encoder layers with eight attention heads ($d_k = 64$) and feed-forward width 2048 process this sequence via scaled dot-product self-attention:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{6}$$

The fused CLS representation is classified through a linear head with dropout 0.3:

$$\begin{aligned} h &= \mathrm{Transformer}(S_0)_{\mathrm{CLS}}, \\ p_k &= \sigma\Big(\big[W_c\ \mathrm{Dropout}_{0.3}(h) + b_c\big]_k\Big), \end{aligned} \tag{7}$$

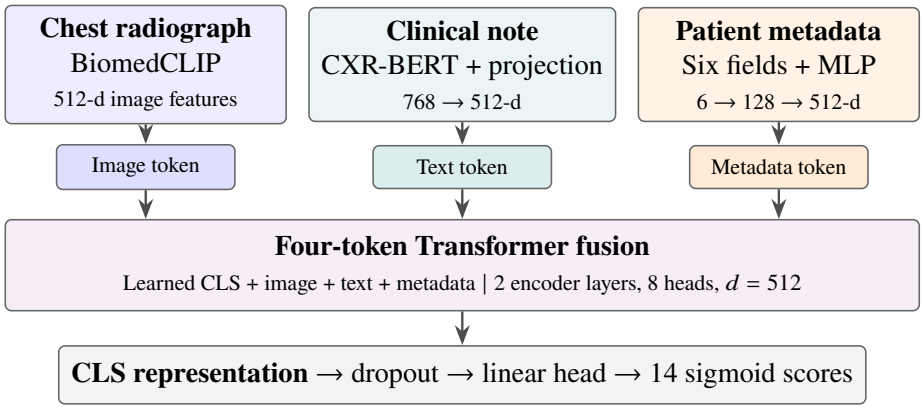


Fig. 1. X-Fusion architecture. Three modality-specific encoders produce 512-d tokens that join a learned CLS token in a two-layer Transformer. The classifier predicts fourteen finding scores. Visual attention and metadata-attribution displays are produced separately by the inference interface.

where $W_c \in \mathbb{R}^{14\times 512}$ and $k = 1, \ldots, 14$ indexes the findings: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia.

3.3 Optimization Objectives

Let p_{ik} denote the predicted probability for finding k in sample i, and $y_{ik} \in \{0, 1\}$ its label. With margin shift $\delta = 0.05$ and $q_{ik} = \min(1 - p_{ik} + \delta,\ 1)$, the Asymmetric Loss per label is:

$$\ell_{ik} = -y_{ik}(1 - p_{ik}) \log(\max(p_{ik}, \epsilon)) \\ - (1 - y_{ik})(1 - q_{ik})^4 \log(\max(q_{ik}, \epsilon)), \quad (8)$$

where $\epsilon = 10^{-8}$. The classification loss averages over the batch: $\mathcal{L}_{\text{ASL}} = B^{-1} \sum_{i=1}^{B} \sum_{k=1}^{14} \ell_{ik}$.

For ℓ_2-normalized image and text features v_i, t_j, the similarity matrix is $C_{ij} = v_i^\top t_j / \tau$ with temperature $\tau = 0.07$. The symmetric contrastive alignment loss is:

$$\mathcal{L}_{\text{NCE}} = \tfrac{1}{2}[\text{CE}(C, a) + \text{CE}(C^\top, a)], \quad (9)$$

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{ASL}} + 0.1\, \mathcal{L}_{\text{NCE}}, \quad (10)$$

where a contains matching in-batch image–text indices.

Table 1. Patient-disjoint dataset partitions. No patient ID overlap exists across splits.

Partition	Images	Patients
Training	21,964	9,939
Validation	2,566	1,104
Test	7,100	1,137
Total	31,630	12,180

Table 2. Positive-label counts by partition. Pneumonia (18 test positives) and Consolidation (62) are extremely rare.

Finding	Train	Val	Test
Atelectasis	3,237	367	879
Cardiomegaly	723	107	387
Effusion	2,687	329	1,270
Infiltration	3,629	433	1,128
Mass	1,600	164	476
Nodule	2,111	226	476
Pneumonia	42	5	18
Pneumothorax	1,148	131	1,024
Consolidation	132	11	62
Edema	395	43	262
Emphysema	557	70	335
Fibrosis	558	57	190
Pleural Thick.	787	94	348
Hernia	63	8	49

4 Dataset and Experimental Setup

4.1 Data Partitioning

The study uses 31,630 images matched from 31,877 available PNG files in a local subset of NIH ChestX-ray14 [7]. The official `train_val_list.txt` and `test_list.txt` define the split boundary. The training/validation pool is further divided by patient ID (seed 42, 10% validation). Table 1 and Table 2 summarize the partition statistics.

Images are resized to 224×224 pixels, converted to RGB, and normalized with ImageNet channel statistics. Training augmentation applies horizontal flips, rotations ($\leq 10°$), and brightness/contrast jitter (0.15). Test preprocessing is deterministic.

4.2 Auxiliary Input Construction and Training Configuration

Age and sex are extracted from the original NIH CSV metadata. In the current implementation, clinical-note text is generated from disease-label templates, and four vital signs (temperature, heart rate, respiratory rate, SpO_2) are sampled from Gaussian distributions conditioned on the presence of Pneumonia or Edema. This creates a direct

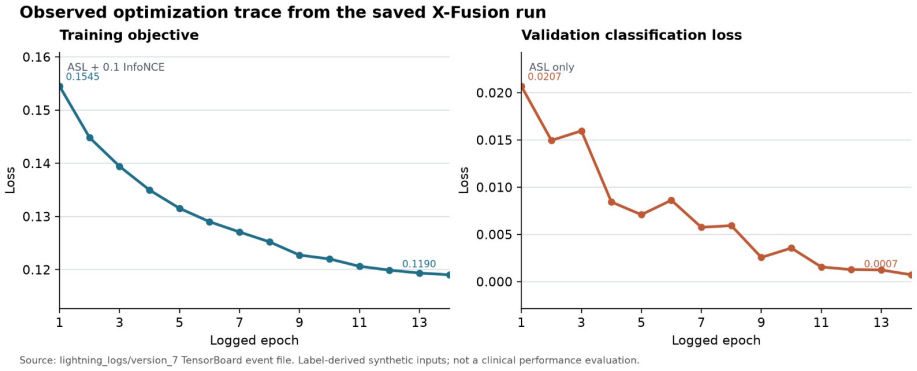


Fig. 2. Optimization traces for fourteen epochs. Training objective includes contrastive regularization; validation uses ASL only. Both operate on label-conditioned auxiliary inputs.

information pathway from target labels to predictors—the core evaluation integrity issue analyzed in Section 5.

Training uses AdamW (lr = 2×10^{-4}, weight decay 10^{-4}), cosine annealing, batch size 4 with 8-step gradient accumulation (effective batch 32), 16-bit mixed precision, and a frozen text encoder. The available checkpoint corresponds to epoch 14 of a 15-epoch schedule.

5 Results and Analysis

5.1 Optimization Records

Fourteen logged epochs show the training objective (ASL + 0.1·InfoNCE) decreasing from 0.1545 to 0.1190 (22.95% relative reduction), and validation ASL decreasing from 0.0207 to 0.0007. Training and validation track different objectives; their scales are not directly comparable. Fig. 2 shows the optimization traces. These curves document convergence of the configured objective, not independent diagnostic performance.

5.2 Recomputed Software Diagnostics

The saved prediction file contains 7,100 distinct image identifiers with fourteen binary targets and fourteen probability columns. We recomputed metrics from these saved outputs: AUROC via empirical ROC integration, average precision via recall-weighted precision sums, and F1 from pooled confusion counts at threshold 0.5. Table 3 reports the results.

5.3 Evaluation Integrity Diagnosis

The near-perfect macro scores are a direct consequence of input construction. The report generator selects disease-specific text templates using the target label list; a model

Table 3. Recomputed diagnostics from saved predictions. These values are inflated by label-conditioned auxiliary inputs and do not constitute clinical performance estimates.

Metric	Value
Macro AUROC	0.9999
Macro average precision	0.9953
Macro F1 (threshold 0.5)	0.9709
Pneumonia precision (0.5)	0.5667
Pneumonia recall (0.5)	0.9444
Pneumonia F1 (0.5)	0.7083

with access to these notes can recover the labels without examining the radiograph. Similarly, the metadata generator samples physiological values from distributions whose parameters change based on Pneumonia or Edema presence. This constitutes *target leakage*: test-case inputs carry target-derived information.

Patient-disjoint splitting prevents image-level or patient-level data leakage between partitions. However, it cannot remedy the within-case dependency where each individual test sample's auxiliary inputs are constructed from its own labels. Consequently, no component of the multimodal performance can be attributed to independently observed clinical context.

For Pneumonia specifically, the saved predictions yield TP = 17, FP = 13, FN = 1 at threshold 0.5 among 18 positives and 7,082 negatives. This illustrates how high aggregate ranking scores can coexist with modest precision on extremely rare labels.

5.4 Independent-Input Reference Experiment

To establish reproducible reference evidence free from auxiliary-input contamination, we trained new models using only independently observed predictors. Four configurations were evaluated:

- **Training prevalence**: constant per-label prediction using training-set label frequencies (chance-level reference).
- **Age + sex**: two recorded demographic features from the NIH CSV.
- **Image pixels**: 16×16 grayscale-resized images (256 pixel features scaled to $[0, 1]$).
- **Pixels + age/sex**: combined 258-feature input.

All non-trivial models use fourteen independent L2-regularized logistic regressions with L-BFGS, regularization strength $C \in \{0.01, 0.1, 1\}$ selected by validation macro AUROC, seed 42, and at most 500 iterations (all models converged). Age values outside $[0, 120]$ are imputed with the training median (51.0). Missing gender is encoded as 0.5. All standardization is fitted on training data only. **No predictor is derived from any disease label.**

Table 4 reports held-out test scores, and Table 5 provides per-label detail for the combined model.

Table 4. Independent-input reference experiment on 7,100 test images. These linear models provide reproducible, leakage-free baselines. They do not validate the tri-modal Transformer.

Configuration	Macro AUROC	Macro AP	Macro F1
Training prevalence	0.5000	0.0695	0.0000
Age + sex ($C = 0.1$)	0.5518	0.0861	0.0000
Image pixels ($C = 0.01$)	0.6317	0.1178	0.0098
Pixels + age/sex ($C = 0.01$)	**0.6344**	**0.1201**	**0.0106**

Table 5. Per-label AUROC for the combined pixels + age/sex reference model (7,100 test images). Best-performing labels tend to have higher prevalence or distinct morphological signatures.

Finding	AUROC	Test positives
Atelectasis	0.640	879
Cardiomegaly	0.749	387
Effusion	0.684	1,270
Infiltration	0.595	1,128
Mass	0.575	476
Nodule	0.633	476
Pneumonia	0.516	18
Pneumothorax	0.646	1,024
Consolidation	0.478	62
Edema	0.718	262
Emphysema	0.552	335
Fibrosis	0.757	190
Pleural Thick.	0.563	348
Hernia	0.777	49

These reference models are deliberately simple linear classifiers on low-resolution features. Their purpose is not to match pretrained deep encoders but to establish a reproducible, uncontaminated measurement floor. Comparing the contaminated X-Fusion scores (macro AUROC ≈ 1.0) directly to these baselines would change both the input space and model family simultaneously—such a comparison does not isolate a fusion effect.

5.5 Explanation Views and Application Interface

The inference interface provides two explanation modalities. **Visual attention**: the final Transformer block's CLS-to-patch attention weights are averaged across heads, reshaped to a 14×14 grid, upsampled, and overlaid on the input image. This is an attention visualization, not Chefer relevance propagation [2] or validated lesion localization. **Metadata attribution**: Kernel SHAP [6] perturbs the six metadata fields while holding image and text features fixed. Absolute contributions are aggregated across outputs and normalized, which removes sign and finding specificity.

A Next.js frontend communicates with a FastAPI inference service. The interface accepts a radiograph, clinical text, and structured fields, returning finding scores with

explanation views. These implement a demonstration workflow and have not been clinically calibrated.

6 Limitations and Future Directions

The primary limitation is target leakage through generated notes and label-conditioned vital signs, which prevents clinical-performance interpretation of the tri-modal scores. Additional limitations include:

- Weak image-level labels from automated NLP extraction [11], not expert annotation.
- Evaluation on a subset (31,630 of 112,120 images) of a single dataset.
- Extremely few positive cases for Pneumonia (18 test) and Consolidation (62 test).
- Uncalibrated decision thresholds not validated on independent clinical data.
- Unvalidated explanation fidelity for both attention overlays and SHAP attributions.

Required future work. (1) Acquire auxiliary inputs (clinical notes, vital signs) recorded independently of the target diagnosis, such as from MIMIC-CXR linked with MIMIC-IV clinical data. (2) Retrain the tri-modal model with corrected inputs; changing test inputs alone introduces distribution shift without repairing training contamination. (3) Conduct matched ablations (image-only, image+text, image+metadata, full fusion) under identical training budgets and evaluation protocols. (4) Select class-specific thresholds on validation data and evaluate the held-out test set once. (5) Report calibration curves, patient-level uncertainty, and external-dataset generalization. (6) Validate explanation fidelity through quantitative localization benchmarks.

7 Conclusion

X-Fusion demonstrates a compact Transformer-based integration of pretrained visual and language encoders with structured patient metadata for fourteen-label chest radiograph analysis, connected to an explanation-enabled web interface. The system's implementation is fully functional and its training records document stable optimization convergence. However, our systematic analysis reveals that label-conditioned auxiliary inputs create within-case target leakage that patient-disjoint splitting cannot remedy, rendering the near-perfect evaluation scores uninformative about clinical discrimination. The newly contributed independent-input reference experiment establishes leakage-free baselines (macro AUROC 0.634) using only real image pixels and recorded demographics. This work underscores that *evaluation integrity verification*—confirming that every predictor is collected independently of the reference standard—is a necessary step before multimodal medical AI systems can claim clinical benefit.

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